

DYLAN BURKEY

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Mechanical Engineering (Concentration in Manufacturing)

GPA: 3.21 (cumulative) and 3.50 (major)

Atlanta, GA

May 2027

EXPERIENCE

Amazon

May 2026 – Aug. 2026

Amazon Robotics Area Manager

Miami, FL

- Collaborated with Reliability and Maintenance Engineering teams to commission new sensor hardware on CARDINAL's conveyor retro-reflective photo eye system; improved induct automation by increasing pallet stacking throughput by 8% and tote FPY by 15%.
- Led cross-functional NPI of a freight management application (Python/JavaScript) to bridge operational visibility into late packages; embedded automated task routing and audits into standard workflow to reduce freight rollover by 11.3k packages/month; implemented in 5 NA FC's.
- Directed 100+ associates daily; coordinated labor and workflow execution to reach operational targets and drive continuous improvement.

Smurfit Westrock

May 2025 – Aug. 2025

Manufacturing Engineering Intern

Mooreville, NC

- Designed and executed test plans across die cutter production runs; analyzed anvil cover wear patterns against die geometry using statistical methods; implemented a data-driven replacement schedule reducing downtime by 25% and extending blade service life by 10%.
- Created and validated engineering drawings and CAD models against design standards for manufacturability; partnered with the mechanical design team using SOLIDWORKS and ArtoisCAD to verify new packing products for production on high-speed converting equipment.
- Implemented an inventory management application to track stock levels, analyze consumption patterns, and flag excess inventory for disposition, reducing stockout-related production downtime by 30% to support maintenance teams.

Snapping Shoals EMC

Aug. 2022 – May 2023

Automotive Engineering Intern

Covington, GA

- Performed mechanical diagnostic/repairs across a fleet of 75+ utility vehicles; managed inventory and procurement to minimize downtime.

PROJECTS

Toothless: Autonomous Robot

Aug. 2025 – Dec. 2025

- Designed and fabricated an autonomous mobile robot integrating machined, 3D-printed, laser-cut, pneumatic, actuating and sensor components with Arduino-based mechatronic control; placed 2nd out of 70+ teams in GT's Creative Decisions & Design competition.
- Applied DFM/DFA and DFMEA through iterative prototyping of components with GD&T specifications in SOLIDWORKS/OnShape.

Pioneer DDJ-FLX10 Rekordbox LCD Display

Feb. 2026 – Present

- Created a custom touchscreen display module for the DDJ-FLX10 to provide an ergonomic Rekordbox interface and seamless navigation.
- Designed and 3D printed a custom enclosure and controller mount in SOLIDWORKS to house a Raspberry Pi LCD and display electronics.

INVOLVEMENT AND LEADERSHIP

Yellow Jacket Aviation

Feb. 2023 – Present

Manufacturing Lead

- Created assembly plans and BOMs for manufacturability of an RV-14 aircraft build; mapped subsystem dependencies to eliminate rework risk.
- Coordinated across engineering, procurement, finance, and operations teams to align build scope with FAA regulations and yearly budget.

Curious Cardinals

Dec. 2023 – Present

Technical Mentor

- Mentored aspiring mechanical engineering students in creating CAD and robotics project portfolios to support professional development.
- Developed a 10-week guide to teach STEM fundamentals and college readiness, currently serving 200+ active students in the country.

American Society of Mechanical Engineers

Sep. 2023 – Present

General Body Member

- Participate in weekly professional development events and networking sessions with engineering companies to enhance industry knowledge.

Kappa Sigma Fraternity

Aug. 2024 – May 2026

Social Chair

- Administered a \$40K semesterly budget, managing vendor negotiations, cash flow tracking, and logistics for 50+ social events for students.

TECHNICAL SKILLS

Software: SOLIDWORKS, Onshape, Creo, ArtoisCAD, Autodesk Fusion, Ansys Granta, Arduino, MATLAB, Minitab

Operational Excellence: Lean Six Sigma, DMAIC, Lean Manufacturing, Project Management, Continuous Improvement, Kaizen, RCA

Hands-On: Mill and Lathe, Bandsaw, Laser Cutting, 3D Printing, Power Tools, Precision Instruments

Fundamentals: Physics/Mechanics, Thermal/Fluid/Heat Transfer, Design/Materials, Computing Techniques, Statistics, Circuitry, Economics

Certifications: Lean Six Sigma Yellow Belt, Amazon AI and Machine Learning Fundamentals, CompTIA A+